Certificate-Gated Neural Proposers for Verifier-Grounded Action Scoring

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Abstract. This paper studies a narrow architecture for settings with exact verifiers: symbolic or replay-certified mechanisms define which candidates are admissible, and small neural components rank, calibrate, reuse, or learn around those certified hypotheses. The main contribution is not that tiny networks solve hard tasks. It is an artifact-feedback loop: rejected verifier rows become reason-coded training signal for later proposers, and repeated zero-false-rejection hypotheses become reusable library priors. The frozen evidence matrix covers certified modular gates, grid-path and delayed-claim stress surfaces, ARC-AGI transform coverage, a tiny-net capacity audit, and Signal replay control. The strongest gate anchor is `prime_gap`: with residue-sieve features removed, the raw scorer reaches 9.13% top-1, a fixed hand-wheel gate reaches 45.12%, and a certified learned-rule gate reaches 68.87% at 100% top-10. The more important substrate rows are C5: rejected-candidate residuals repair unsafe fixed-budget proposer rankings, and certified hypothesis-library priors improve tight-budget proposal coverage under a held-out split. ARC is an evidence surface for priors and failure taxonomy; Signal is application pressure under replay, shadow logging, and a guarded live-active pilot. The paper explicitly does not claim broad ARC competence, comparative live neural-control improvement, or general end-to-end task solving.

1. Introduction

Exact verifiers are common in domains where action correctness can be checked algorithmically even when discovery cannot: theorem proving, program synthesis, combinatorial search, formal shortest-path problems. The dominant style in modern learned policy work treats the verifier as a reward signal for an end-to-end neural policy. We argue this is the wrong default when the candidate space contains many structurally impossible actions that the verifier can cheaply eliminate.

Our thesis is the **certificate-gating loop**: a learned scorer should consume the **survivors** of a verifier-checked symbolic filter, and the filter itself can be produced by a rule proposer gated by a soundness-certificate verifier. The scorer ranks the residual; the gate eliminates the rest; nothing the gate emits can violate correctness because every rule it accepts carries a deterministic proof.

The paper's experiment-family set is now frozen. New domains and new benchmark families are roadmap items, not body evidence, until the matrix in Table 1 is deliberately revised. The frozen families are: certified modular gates, rejected-candidate learning, certified hypothesis-library feedback, tiny-net capacity auditing, ARC evidence surfaces, and Signal replay pressure. Grid-path and delayed-claim remain supporting stress surfaces because they test whether the same candidate/verifier/scorer/trace contract survives outside the prime-gap anchor.

The two main empirical contributions are:

(M1) **Rejected-candidate learning.** Failed verifier proposals are not discarded. They are logged with reason codes, false-rejection counters, split keys, seed intervals, and marginal verifier utility, then reused as targets for a safety-first residual ranker. This repairs the fixed-budget modular proposer failure without giving the neural model authority over correctness.

(M2) **Held-out certified-library feedback.** Repeated zero-false-rejection hypotheses are promoted into a library, converted into slot priors, and evaluated on held-out tasks under the same verifier

budget. This is the cumulative-learning direction: certified patterns become reusable proposal structure rather than one-off fixture wins.

The older axis-style claims remain in the paper as support and boundary evidence. Certified gates reduce the residual action space; the fixed-budget neural proposer result is negative and becomes a limitation; verifier-RL calibration is a prime-gap-only support row; tiny-net capacity results constrain architecture claims. Those axes are not new experiment families. They are checks on the same loop.

Operationally, the certificate-gating loop yields an artifact-feedback mechanism that recycles verifier outputs into future gates, priors, and ranking targets (Table 2), and a generalized task contract that keeps the same candidate/verifier/scorer/trace interface across domains from arithmetic to replay-grounded control (Table 3).

The evidence is heterogeneous by design, so the paper uses a single protocol matrix rather than a single leaderboard. The primary question in every row is not "how smart is the small network?", but "what does the environment supply, what does the verifier own, what is the network allowed to rank, and what reusable artifact is produced?"

Family	Candidate Set	Verifier	Certificate / Gate	Scorer Role	Trace Artifact	Result	Claim	Limitation
prime_gap	gap-gap actions plus certified modular rule filters	segmented-sieve replay and modular certificate checks	zero-false-rejection divisibility certificates over train/validation support	rank residual gaps and budgeted certified rules	crlp.compare.v1; crlp.rule_trace.v1; crlp.rule_search.summary.v1	learned gate 68.87% top-1 vs 45.12% handwheel and 9.13% raw	certified modular gates shrink the residual action space	static neural proposer is not superior to deterministic baselines at fixed budget
grid_path	path moves plus symbolic local/topology/bound predicates	BFS shortest-path successor and rule certificate checks	legality, trap, clear-path, and distance-dominance certificates	rank legal residual moves after symbolic constraints	crlp.compare.v1; crlp.symbolic_bound_domain_transfer.v1	learned gate lifts priors-enabled raw 75.80% to 90.10%	same gate contract works outside residue arithmetic	supporting transfer surface, not a main contribution
delayed_claim	claim actions, threshold gates, and utility-margin rules	deterministic hidden-outcome simulator and rule certificate checks	finite-generator safety check plus false-rejection counters	rank utility only after safety dominates the objective	crlp.delayed_claim_rule_trace.v1; crlp.rejected_candidate_learning.v1	threshold+learned gate reduces four actions to one; noisy rows expose failures	reason learning needs domain-aware safety constraints	reason-prior transfer alone can fail safety; claim remains domain-scoped
rejected_candidate_learning	rejected candidate rows grouped by task, split, reason family, and seed	original rule/certificate checker that rejected or accepted each row	false-rejection-first budget filter over reason priors and residual scores	learn residual ordering from rejection reasons and marginal verifier utility	crlp.rejected_candidate_learning.v1	learned residual reaches 7 accepted rules / 245 utility with zero false rejections	verifier failures become training signal	not a generic reason-code ranker; delayed-claim safety remains scoped
hypothesis_library_feedback	certified hypothesis templates, slot bindings, and compiled programs	train-only usefulness, type checks, and held-out fixture split evaluation	zero-false-rejection promotion before a binding enters the library	rank promoted slot priors and residual program candidates under tight budgets	crlp.hypothesis.evidence.v1; crlp.hypothesis_library_feedback_result.v1	slot priors raise cyclic fill@1 from 2 to 4; held-out split rows = 30	certified learning can accumulate through reusable library priors	public ARC promotion remains scoped evidence, not a headline ARC score
multitask_tiny_net	shared residual ranking rows across prime/grid/claim/ARC	task-local labels and certified downstream gates	per-task held-out metrics, OOD ARC variants, calibration bins	shared residual scorer with domain-specific diagnostics	crlp.multitask.summary.v1; crlp.multitask.tiny_shared_net.v1	ARC-1; OOD shared-plateau 5.57pp; capacity is not monotonic; plateau audit is underpowered	representation/protocol limits dominate naive width scaling	architecture-size claims stay gated by the plateau audit
arc_agi_evidence_surface	cell-color-mask transforms, programs, and hypothesis-derived candidates	rule-fact consistency, unique compiled prediction, and public corpus loader	program/hypothesis certificate rows and scorer-invariant loader summaries	rank bounded candidates; scorer does not own correctness	crlp.arc_loader.summary.v1; crlp.program_proposer.summary.v1; crlp.hypothesis.evidence.v1	public eval coverage is 7/400 tasks and 1057 cells under the measured surface	ARC exposes priors, grammar coverage, and failure taxonomy	not a broad ARC competence or benchmark-score claim
signal_flight	certified control-target hypotheses plus nine WASD action candidates	offline counterfactual simulator replay, live shadow frame comparison, and live active telemetry	replay utility, regret, damage/death counters, shadow disagreement logs, and live safety gates	rank bounded control candidates beside heuristic control	crlp.signal_flight.summary.v1; crlp.signal_shadow_replay_audit.v1; crlp.signal_live_active.v1	Signal top-1 79.76% vs baseline 67.86% (+11.90pp), utility 0.1432 vs 0.1359, regret 0.0073 vs 0.0146, damage 2.38% vs 2.38%, deaths 0.00% vs 0.00%; live shadow frames=22191. Signal live neural-active pilot executed 250 recent neural-source rows from checkpoint f12b644ed9fa with p95 inference 0.124 ms, 249 replay-ready rows, safety-gate rows=250, damage events=0, deaths=0; no baseline comparison is claimed.	application pressure for replay-grounded residual ranking and guarded live neural-active execution	active neural versus heuristic performance is not yet claimed

Table 1: Frozen evidence matrix. Every result row uses the same structure: domain/data, grammar, candidate set, verifier/certificate, scorer role, trace artifact, result, takeaway/claim, and limitation. The paper may tighten these rows, but it should not add new experiment families without revising this matrix first.

Non-claims. The paper does not claim general task solving, competitive ARC performance, comparative live neural-control improvement, or that small networks replace scale. It does not claim that a generic reason-code ranker is safe across domains, or that wider MLPs solve the ARC geometry failure. It claims a narrower protocol result: when certified candidate mechanisms own admissibility, small learned components can usefully rank residual candidates, calibrate decisions, reuse certified hypotheses, and learn from verifier rejections under an auditable trace contract.

The artifact now emits `crlp.experiment_schema.v1` rows for every certified task with `make experiment-schemas`, plus shared protocol rows for candidate rows, verifier outcomes, ranker decisions, discovered rules, and library promotions. The paper should be read against that schema: task-specific verifiers and grammars change, but candidate row shape, verifier-outcome shape, leakage discipline, proposer baselines, trace replay, and verifier-RL comparisons are shared.

2. Related Work

2.1. Verifier-guided learning and process supervision

DeepSeek-R1 is the closest contemporary confirmation of the verifier-first premise at frontier scale: DeepSeek-AI report reinforcement learning from verifiable rewards (RLVR), where math, code, and format checks provide correctness signals without a learned reward model [1]. That supports the architectural direction of this paper, but it is not the same artifact. DeepSeek-R1 couples verifiable rewards to massive language-model sampling and opaque training infrastructure; this work keeps the verifier line auditable by putting candidate enumeration, certificate checks, accepted rules, rejected proposals, and scorer metrics into schema-attested C11 traces. The contrast with the standard learned-reward line [2], [3], [4] is therefore not "RL is unnecessary"; it is narrower: when an exact verifier exists, learned components should rank or propose within a verifier-bounded search, not replace the correctness oracle.

The learned-verifier line starts from a different point. GSM8K-style verifier-guided decoding trains a model to rank sampled solutions after generation [5]; process-supervision work trains reward models on intermediate reasoning steps rather than only final outcomes [6], [7]; V-STaR adds a self-training loop in which both correct and incorrect sampled solutions improve the verifier [8]. Those systems are natural relatives because they all use a learned component to reduce search over candidate reasoning traces. The boundary in this paper is that the verifier is not learned: verifier outputs may provide labels, rewards, certificates, and diagnostics, but accepted gates require deterministic certificates and scorer inputs are checked against a typed leakage policy. In short, learned verifiers ask "which sampled answer should we trust?"; this work asks "which candidates remain after a sound filter, and how should we rank that residual?"

The regularization used in Axis 3 is also borrowed, not claimed as novel. RLHF systems made KL-to-prior and entropy-style stabilization standard for preference rewards [3], [4], with PPO and trust-region policy optimization as the broader policy-gradient background [9], [10]. Our contribution is to show the same stabilization pattern under exact verifier rewards: strict policy-gradient updates inflate confidence on the prime-gap modular gate instance, while the KL/entropy branch preserves ranking and improves calibration. Calibration itself is reported in the standard ECE/temperature-scaling frame of [11].

2.2. Neuro-symbolic search with formal checkers

AlphaGeometry is the closest structural ancestor of this work. It interleaves a neural language model that proposes auxiliary geometric constructions with a symbolic deduction engine that verifies whether the construction completes the proof [12]; AlphaGeometry2 extends that line to stronger olympiad-geometry performance [13]. GPT-f is the earlier theorem-proving version of the same pattern: a language

model proposes proof steps and a formal system checks them [14]. Our architecture is deliberately smaller and more explicit. The grammar is task-local, the proposer sees structural rule descriptors rather than a full natural-language proof context, and every accepted rule is emitted into a trace with support, marginal utility, certificate status, and false-rejection counts. The comparison is therefore not scale; it is audit posture. Their systems show that neural proposal plus formal checking is powerful at large scale; this paper asks how much of that pattern can be made schema-attested, reproducible, and heterogeneous across small verifier-equipped tasks.

2.3. Program synthesis and library reuse

DreamCoder and Hypothesis Search are the closest relatives of the hypothesis layer rather than the modular rule gate. DreamCoder learns reusable DSL abstractions through wake-sleep Bayesian program learning [15]; Hypothesis Search samples natural-language hypotheses, compiles them into executable programs, and keeps the train-consistent ones [16]. `NNHypothesis` sits between those two poles: templates and slot types are deterministic, but bindings are induced from task evidence and compiled into canonical `NNProgram` rows that the same certificate/proposer machinery can score. This is why the ARC and `program_synthesis` evidence rows matter: they show the substrate's "hypothesis -> program -> certificate -> proposer summary" path outside the current modular arithmetic grammar.

The library-feedback claim is narrower than open-ended program induction. A binding enters the library only after a verifier accepts it with zero false rejections, and the promoted binding is evaluated as a prior under a held-out budget. This makes the connection to program-synthesis systems explicit without turning the paper into a new DSL benchmark.

2.4. ARC abstraction benchmarks

ARC is used here as one hard abstraction substrate in the evidence portfolio. Chollet's ARC framing emphasizes skill-acquisition efficiency and the need to account for priors [17]. The 2024 ARC Prize report and recent test-time-training work show the current leaderboard direction: large-model or program-synthesis systems with per-instance adaptation achieve far higher ARC scores than this artifact [18], [19]. Our public ARC result should therefore be read narrowly: public ARC evidence surface improves from v10 eval 4 tasks/187 cells to v15_measured eval 7 tasks/1057 cells; v15_measured train remains 24 tasks/1187 cells. The fixed `0520fde7` transform gate gives a clean scorer-vs-certificate contrast on one task. The research claim is not ARC performance; it is that feature provenance, certificate-gated transforms, matched proposer budgets, and replayable evidence rows make the priors and failure modes visible.

3. Method

3.1. The Certificate-Gating Loop

The reusable architecture is:

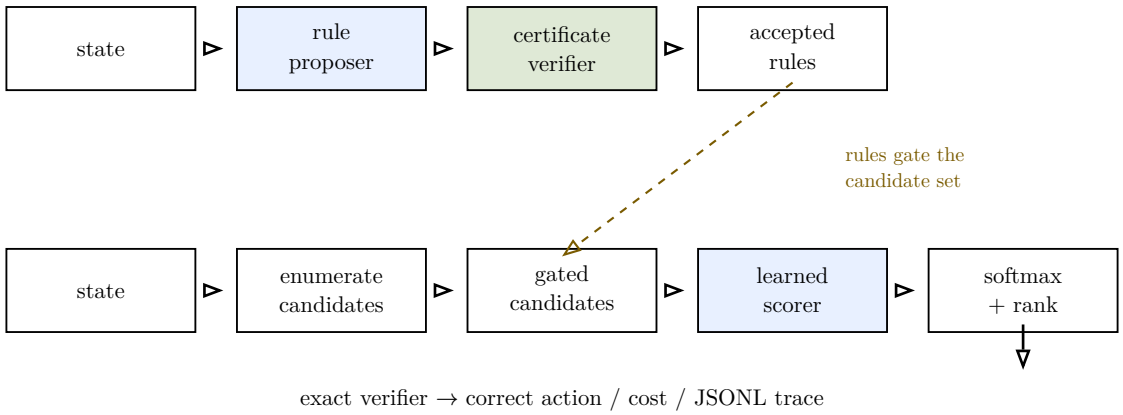


Figure 1: The certificate-gating loop. A rule proposer (top) produces candidate rules that a soundness-certificate verifier accepts or rejects. Accepted rules deterministically prune the candidate set; a shared neural scorer ranks the survivors; the exact verifier supplies the correctness signal.

The design invariant is **the model proposes, the verifier decides, the trace remembers.** A rule r is accepted only if a deterministic algebraic check certifies that r cannot reject the true action on any state in the support set, so soundness does not depend on the proposer being well-trained.

Verifier Output	Trace Artifact	Reusable Form	Next Ranking Use
accepted rule	rule trace with support, certificate status, and false rejections	gate program	prune impossible candidates before scoring
rejected candidate	reason-coded verifier rejection row	unsafe / no-support / duplicate prior	avoid low-utility proposals at fixed budget
certified hypothesis	typed program evidence row	library prior over template slots	promote reusable transforms under a verifier budget
replay outcome	counterfactual utility, damage, and death row	bounded action utility target	rank control candidates without owning safety

Table 2: Artifact feedback loop. The paper's central mechanism is not raw neural accuracy. The environment produces certified or replayed artifacts; those artifacts become gates, priors, and ranking targets for the next proposal pass.

3.2. Generalized Task Schema

Every verifier-backed experiment in the artifact emits the same schema row:

```
Task T = {
  state distribution S_T,
  candidate enumerator A_T(s),
  observable features x_T(s, a),
  neural residual scorer f_theta(x_T),
  exact verifier V_T(s, a),
  symbolic grammar G_T,
  certificate checker C_T(rule),
  rule proposer q_phi(rule | structural(rule)),
  gate executor R_T(rule, s, a),
  trace/replay/RL protocol
}
```

Task	Candidate Set	Verifier	Certificate / Gate	Scorer Role	Trace Artifact	Protocol Point
prime_gap	candidate gaps + modular rule candidates	segmented sieve next-prime verifier	divisibility certificate; zero false rejection	rank residual gaps and budgeted rules	crlp.compare.v1, crlp.rule_trace.v1	certified modular gate search anchor
grid_path	four moves + legality/distance rules	BFS shortest-path successor	move cannot reject optimal successor	rank legal residual moves	crlp.compare.v1	same gate contract outside prime residues
delayed_claim	trust / ignore / request / cautious	deterministic hidden-outcome simulator	finite-generator safety check	rank actions under delayed labels	crlp.trace.v1 replay rows	trace replay plus production threshold+margin gate
arc_agi	cell colors + transform/program candidates	held-out grid and train-pair transform check	train consistency + unique compiled prediction	rank cells, programs, and hypothesis-derived transforms	crlp.arc_transform.summary.v1, crlp.hypothesis.evidence.v1	abstraction task exposes priors and grammar coverage
program_synthesis	list/int DSL abstractions	run candidate abstraction on examples	monomorphic DSL type check	rank reusable typed programs	crlp.hypothesis.evidence.v1	same hypothesis → program → certificate path beyond ARC
signal_flight	nine WASD control candidates	authoritative Signal simulator replay + live active telemetry	counterfactual utility + damage/death replay + safety-gate log	rank bounded real-time control actions	signal.flight_counterfactual.v1, crlp.signal_flight.summary.v1, crlp.signal_live_active_summary.v1	application of same gate/scorer for replay-grounded control

Table 3: The task instantiations share the same verifier-gated action-scoring contract. Each row repeats the same question: what is the candidate set, which checker owns correctness, what becomes a certificate, what the learned model is allowed to rank, which trace records the decision, and what the task proves about the protocol.

The leakage rule is shared: verifier outputs may be labels, rewards, certificates, diagnostics, or executed gates, but they are not default scorer inputs. Proposer inputs are structural rule descriptors; support-derived marginal utility labels and false-rejection diagnostics are targets, not features. In the artifact this is a typed contract, not just prose: every scorer feature declares provenance (`state_observation`, `structural_prior`, `verifier_derived`, or `derived_from_gate`), every task declares a leakage policy (`forbidden`, `priors_allowed`, or `legacy`), and schema emission fails if enabled scorer features violate that policy.

3.3. Trace and Checkpoint Contract

Every step writes a JSONL row with schema `crlp.trace.v1`, including a stable per-action identifier, the candidate scores and probabilities, the selected and target indices, the verifier result, and verifier cost. Replay resolves `target_index` → stable `action_id` → regenerated candidate row, so reordering action ids cannot silently change replay semantics. Checkpoints carry SHA-256 model identity, a `parent_model_hash`, a `feature_encoder_version`, and a `feature_set`; comparison reports (`crlp.compare.v1`) carry both hashes, the evaluated `feature_set`, and machine-readable reliability bins.

4. Task Family: Certified Modular Gate Search

The prime-gap task is the current production instantiation of certified modular gate search. The family is defined by the proof shape, not by primes: enumerate arithmetic candidates, propose residue/period/divisibility rules from structural descriptors, accept only rules with deterministic certificates, gate impossible candidates, and rank the residual with the learned scorer. Prime constellations, k-free or polynomial-value filters, and Fibonacci/Lucas recurrence gates are natural next members because they preserve that modular certificate contract while changing the arithmetic substrate.

The artifact now includes a smoke implementation of those next members in `examples/modular_gate_search.c`. It emits `crlp.modular_gate.summary.v1` rows for `prime_constellation`, `kfree_squarefree`, and `fibonacci_lucas` under matched rule budgets. Each task reports `grammar_order`, `random`, `structural_heuristic`, `learned_structural`, `train_marginal_oracle`, and `validation_marginal_oracle` rows. The learned proposer is a structural-only linear model trained on train-set rule coverage labels, and the smoke requires zero held-out false rejections. These rows defend the family boundary; they do not yet replace the production `prime_gap` learned-gate table.

4.1. Prime-Gap Setup

The state is a prime p; the candidate space is gaps $g \in \{2, 4, ..., 256\}$ plus an overflow action; the verifier returns the true next-prime gap. The default candidate encoding is the minimal feature set $[\log p, \text{norm_gap}, \text{overflow_flag}]$. A full-sieve feature set that adds precomputed divisibility flags for $p + g$ modulo small primes is retained only as a leakage ablation. The scorer is a shared MLP returning one scalar per candidate; softmax over candidates produces the proposal distribution.

The default split is $p < 800,000$ train, $800,000 \leq p < 1,000,000$ validation, $1,000,000 \leq p \leq 3,000,000$ test; all reported numbers use 30,000 held-out test states unless stated otherwise.

4.2. Why this domain

- A fast deterministic verifier at all scales (segmented sieves, wheel filters, block proofs).
- A compact totally-ordered candidate space.
- An exhausted classical literature — we cannot beat a sieve, which sharpens the question to "what does the learned component contribute under perfect feedback?"
- Rich algebraic structure for the rule grammar (residue classes, divisibility certificates).

5. Axis 1: Certified gates reduce the residual action space

The same trained prime-gap scorer inside the certified modular gate instance is evaluated under three gate conditions on the canonical 30,000-sample held-out benchmark. Top-1 results:

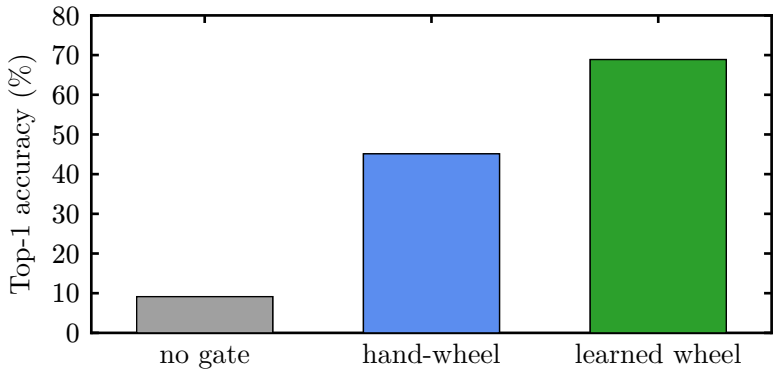


Figure 2: Top-1 accuracy by gate on the canonical 30,000-sample held-out benchmark. The clean raw-to-learned delta is **9.13 → 68.87**. The fixed-hand-wheel-to-learned delta is **45.12 → 68.87** (+23.8pp).

The former 45.12% raw-scorer result is now a full-sieve-feature leakage ablation, not the headline.

Full metrics:

Gate	Top-1	Top-10	NLL	ECE
no gate	9.13%	78.01%	2.9994	0.1076
hand-wheel	45.12%	99.77%	1.5902	0.0834
learned wheel	**68.87%**	**100.00%**	**1.0056**	0.1402

Table 4: Gate ablation, minimal-feature scorer, identical weights across rows. Timing fields are excluded from the canonical comparison because they vary by runner; the numerical claim here is the accuracy and calibration effect of certified pruning.

6. Axis 2: Rule search must beat fixed-budget symbolic baselines

The canonical rule grammar is `reject candidate n when n > m and n % m == r`; the certificate verifier accepts a rule iff $\gcd(m, r) > 1$ (so n is necessarily composite). The matched-budget baseline table also reports conditional gap filters (`if p % m == a then reject gap % m == b`) because that grammar is less saturated and exposes proposer failure more clearly. Zero false rejections is a structural property of the certificate. The proposer input is structural-only: condition kind, moduli, reciprocal scales, residues, zero-residue indicators, gcd ratios, and derived candidate residue. Empirical support and false-rejection counts are diagnostics, and budgeted marginal utility is the neural target, not a proposer feature.

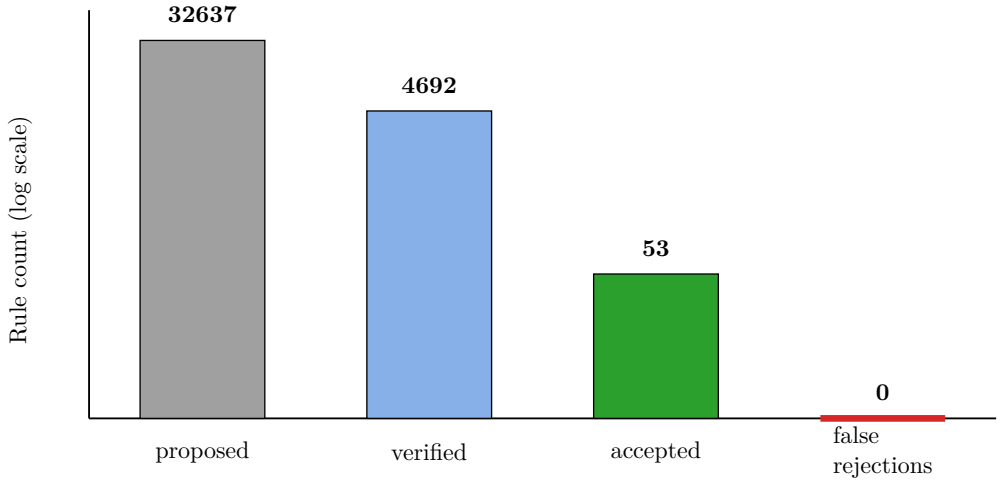


Figure 3: Rule-search acceptance funnel. Of 32,637 proposed rules, 4,692 pass the soundness certificate, 53 survive support and false-rejection screens, and **zero** falsely reject the true next-prime gap. The accepted set recovers 79.79% of wheel-equivalent verifier work.

Grammar	Rule ordering	Budget	Exact probes	Saved
candidate	neural structural	16	1.98	72.49%
candidate	random, seed 20240511	16	3.86	46.30%
candidate	grammar order	16	1.93	73.22%
candidate	structural heuristic	16	1.93	73.22%
candidate	validation-marginal oracle	16	1.93	73.22%
conditional gap	neural structural	16	7.04	2.15%
conditional gap	random, seed 20240511	16	6.94	3.49%
conditional gap	grammar order	16	3.20	55.57%
conditional gap	structural heuristic	16	3.20	55.57%
conditional gap	validation-marginal oracle	16	3.11	56.83%

Table 5: Matched 16-rule-budget proposer baselines over the same split and certificate verifier. Lower exact probes is better.

What this evidence supports. "Certificate-gated structural rule discovery over this grammar produces a sound, useful rule set without support-count feature leakage." The soundness comes from the certificate; the coverage comes from the grammar plus accepted certified filters.

What this evidence does not support. "The neural proposer, specifically, is essential." Table 5 is a negative result for that stronger claim: simple deterministic orderings beat the neural structural proposer at the fixed 16-rule budget on both reported grammars, and the conditional gap grammar exposes a larger failure. The next neurosymbolic experiment must use a richer grammar and show that learned ordering beats grammar-order and structural-heuristic baselines under the same budget.

This negative row is the hinge between the gate result and the artifact-feedback result later in the paper. A plain neural proposer trained only on structural descriptors is not enough; the system becomes useful when verifier failures are promoted into reusable reasons. The generated C5 reason-learning row makes that repair measurable: at budget 8, `model_score` selects 7 unsafe rules and causes 172 validation false rejections, while reason-prior and trained-residual orderings recover useful certified rules with zero validation false rejections (Figure 8). Thus the paper does not claim "neural proposal wins"; it claims that verifier-grounded traces can turn failed proposals into the next proposal policy's training signal.

7. Axis 3: Verifier RL must preserve calibration

Starting from a single supervised checkpoint, we clone two RL branches: strict policy gradient on exact rewards, and the same gradient regularized with KL-to-supervised-prior + entropy.

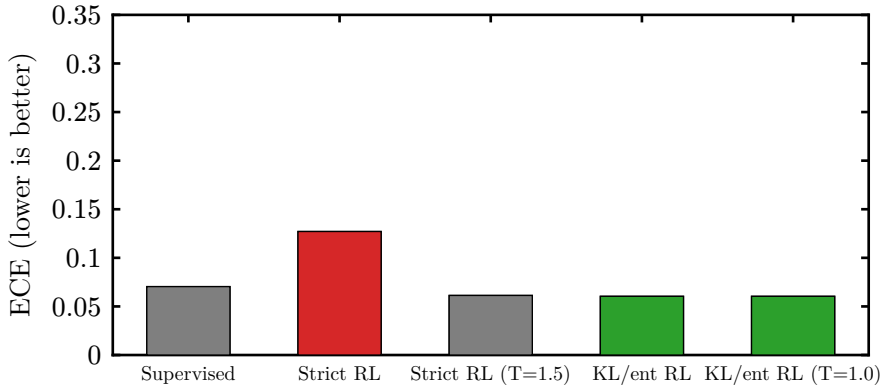


Figure 4: Expected calibration error by RL variant. Strict verifier RL degrades calibration (ECE 0.070 $\rightarrow$ 0.127) while leaving top-1 **unchanged at 9.13%** across all variants. Post-hoc temperature scaling needs $T = 1.5$ to recover; KL/entropy regularization fixes it at training time without that crutch.

Policy	**T**	**Top-1**	**Top-10**	**NLL**	**ECE**
Supervised	1.00	9.13%	78.01%	2.9170	0.0704
Strict verifier RL	1.00	9.13%	78.01%	3.0611	**0.1271**
Strict RL, temp-scaled	1.50	9.13%	78.01%	2.9063	0.0613
KL/entropy RL	1.00	9.13%	78.01%	2.9056	0.0604
KL/entropy RL, temp-scaled	1.00	9.13%	78.01%	2.9056	0.0604

Table 6: RL ablation, single cloned supervised checkpoint, default split.

The mechanism: strict policy gradient on a sparse exact reward signal pushes logit magnitudes without changing their relative ordering. The KL penalty bounds movement away from the reference policy; the entropy bonus discourages collapse to a single overconfident action. Together they reproduce the standard RLHF training stabilization, applied here to verifier rewards instead of preference rewards.

7.1. Reliability Diagram

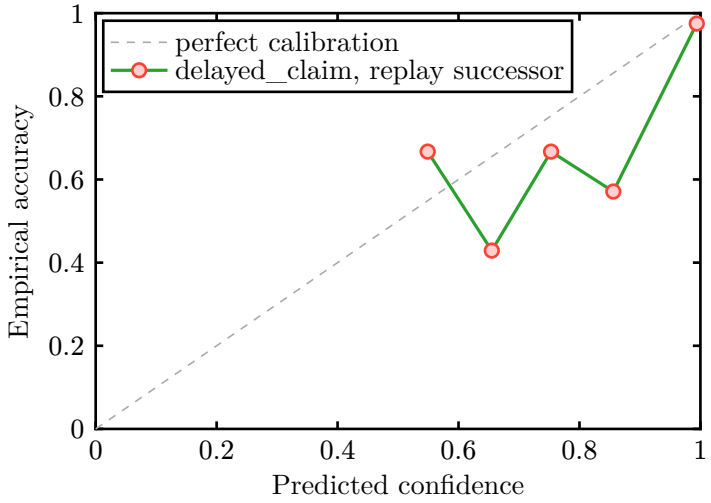


Figure 5: Reliability diagram for the trace-replay successor on `delayed_claim`. Bins from `results/latest/comparison.jsonl`. The high-confidence bin (894 of 1000 samples, $\hat{p} = 0.994$, $\hat{p}_{\text{emp}} = 0.975$) carries most of the mass; the low-population mid bins are noisier.

8. Cross-Task Learning

The same shared neural library, scorer architecture, and verifier-backed task contract instantiate across the task families in Table 3. This section reports how the learned component behaves on each. The reported numbers are produced by `make results` and live in `results/latest/comparison.jsonl`; the figures in this section are driven by `paper/data.typ`, autogenerated from those same artifacts by `paper/scripts/extract_learning_curves.py` so the figures cannot drift from the underlying runs.

8.1. Per-task gate ablation

Different tasks instantiate the certificate-gating loop differently: the prime-gap modular gate instance and grid-path have **gate ablations** in the strict sense (same scorer, with vs without the gate); delayed-claim has a trace-replay successor plus a production threshold+margin gate; ARC-AGI has a **scorer-vs-gate** ablation where the two rows are different architectures (a neural cell scorer and a symbolic transform gate over the same task). A single common-axis bar chart would obscure those distinctions, so Figure 6 reports four small per-task panels with row labels that name what is being compared.

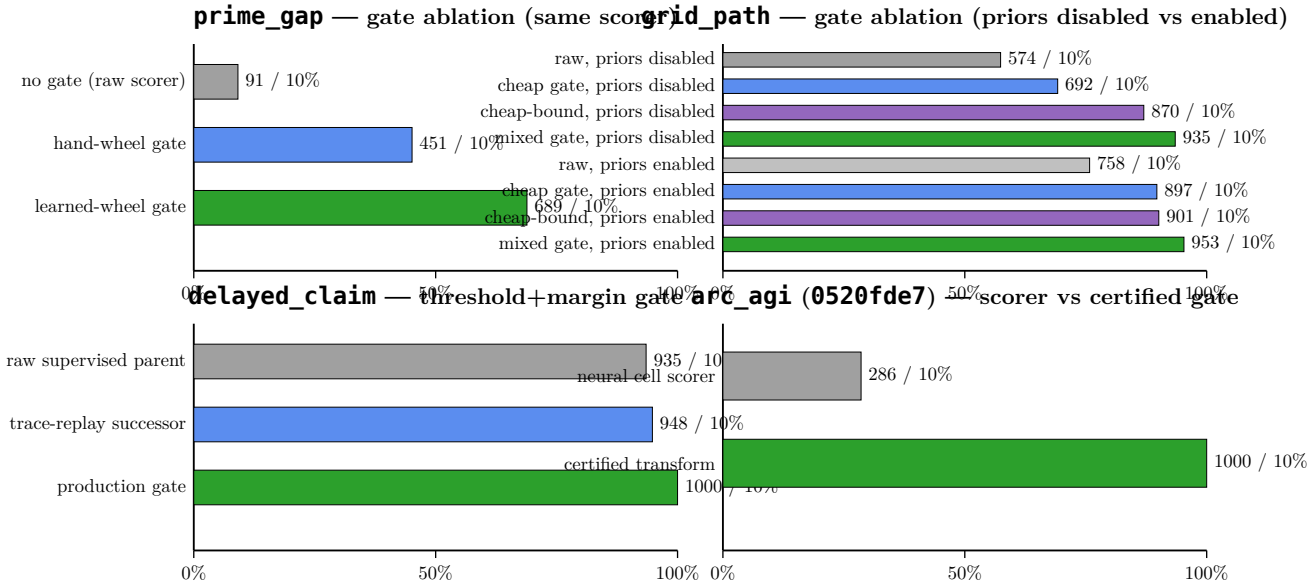


Figure 6: Per-task accuracy by ablation row. Each panel labels its own comparison so the rows are read against the right baseline: the prime-gap modular gate instance and grid-path are **gate ablations** (same scorer, with vs without the gate; grid-path additionally varies whether the Manhattan structural priors are enabled in the scorer features); delayed-claim includes the trace-replay successor and the production threshold+margin gate over the same generator, with parent-hash chain attesting model identity; ARC-AGI is a **scorer-vs-gate** ablation where the two rows are different architectures over the same `0520fde7` task, not the same model with vs without a gate. The ARC `100.00%` is on nine held-out cells of one task; the public ARC evidence surface is the generated v15 measured row reported in public ARC evidence surface improves from v10 eval 4 tasks/187 cells to v15_measured eval 7 tasks/1057 cells; v15_measured train remains 24 tasks/1187 cells., with the hypothesis-induction layer in Section 10 adding transform families beyond the older v10 floor (zero baseline-rate, 4/4 induction-rate on the synthetic suite under `crlp.arc_induction.parity.v1`).

8.2. Training curves per task

The four tasks expose learning on different timescales. `grid_path` and `delayed_claim` emit per-step training traces under `--trace-train`, so we plot rolling 50-step accuracy across a 4000-step supervised run. `prime_gap` does not emit a per-step training trace; its first-class binary reports per-epoch super-

vised loss on stdout, which decreases over 8 epochs and is what we plot. `arc_agi` trains its cell scorer in a single supervised block and exposes per-epoch eval only by re-running the binary with different `--epochs` values, so we plot the top-1 / NLL curve over a 4-point sweep `{1, 2, 4, 8}`. Each panel labels its own x-axis honestly. The plotted data is imported verbatim from `paper/data.typ`, autogenerated by `paper/scripts/extract_learning_curves.py`.

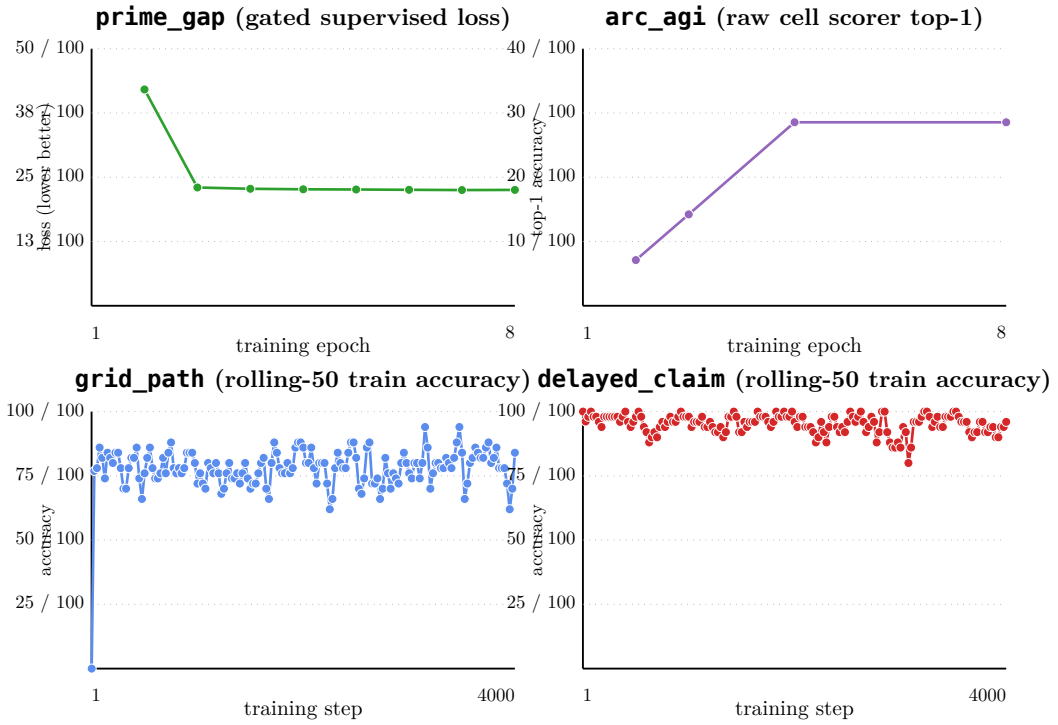


Figure 7: Per-task training-time curves on four different timescales, each labeled with its own y-axis: **`prime_gap`** plots the per-epoch supervised loss of the **gated** scorer (learned-wheel supervised), which decreases from 0.42 to 0.23 over 8 epochs and plateaus around epoch 3 — the raw scorer's top-1 is structurally flat at ≈12% on this minimal feature set because the 129-way candidate space exceeds the residual signal, which is the whole point of the gate. **`arc_agi`** plots per-epoch top-1 of the raw cell scorer on `0520fde7` across the sweep $\{1, 2, 4, 8\}$ epochs; the scorer climbs from 7.11% to 28.56% over the first four epochs and then plateaus, after which additional epochs only inflate NLL (visible as ECE drift in Figure 6). **`grid_path`** and **`delayed_claim`** plot rolling 50-step accuracy across the 4000-step supervised training trace produced under `--trace-train`. All four series are imported from `paper/data.typ`, autogenerated by `paper/scripts/extract_learning_curves.py` from the binaries' own outputs.

8.3. Per-task interpretation

Prime-gap modular gate instance. Figure 2 already shows the **9.13 → 45.12 → 68.87%** progression at final evaluation. The training curve in Figure 7 shows that the cell scorer alone reaches ≈20% top-1 — well below the hand-wheel and learned-wheel gated rows. The model is learning **something**, but on a 129-way action space the gate is the load-bearing component.

Grid-path. The model has access to Manhattan structural priors (`candidate_manhattan_to_goal`, `candidate_manhattan_delta`); the **prior-disabled** rows quantify their value. With priors enabled, the raw scorer is at 75.80%; the cheap-local gate lifts this to 89.70% (+13.9pp), the cheap-bound gate reaches 90.10% (+14.3pp), and the mixed topology+bound+exact-distance gate reaches 95.30% (+19.5pp). With priors disabled at evaluation time, the cheap-local gate lifts 57.40% -> 69.20%, the

cheap-bound gate reaches 87.00%, and the mixed gate reaches 93.50%. The clean cross-domain gate claim is now the non-exact cheap-bound family; the distance-dominance rule remains an explicitly labeled oracle-style upper bound. The family/budget proposer sweep is a robustness check: the default neural selector now targets train/validation residual marginal utility, which prevents overlapping clear-path rules from consuming the fixed budget, but it is not an "always wins" result at every tiny budget and seed.

Delayed-claim. The supervised parent reaches 93.5% accuracy and ECE 0.0363; the trace-replay successor improves to 94.8% / ECE 0.0166 over the same evaluation set with a parent-hash chain attesting identity. The production threshold+margin gate then certifies 4 rules for the neural, closed-loop, structural, and validation-oracle orderings, with zero held-out false rejections. The accepted threshold rules remove dominated `ignore`/`act_cautiously` actions; the accepted utility-margin rules resolve the hard `trust` versus `request_more_evidence` residual. On the held-out generator this leaves 1.00 candidate/action, so the gated row is 100.0% accurate with NLL 0.0000 and ECE 0.0000. The same direct gate stays exact on a shifted seed band, but clean-observation soundness is not the same thing as noisy-observation robustness: with observation noise 0.03 it records 23/1000 false rejections, and on a hard uncertainty+noise band it records 104/1000. The indirect risk-margin grammar is safer but less decisive: the structural/oracle row has zero false rejections under observation noise 0.03, but only 94.6% accuracy and 1.35 candidates. The reliability diagram in Figure 5 is from the replay successor before this deterministic gate.

ARC-AGI. On the fixed `0520fde7` task, the neural cell scorer reaches 28.56% top-1 with **ECE 0.5574 — substantially worse calibration than any of the other three tasks**. The cause is structural: the cell scorer must distribute probability mass over 10 candidate colors per output cell while seeing only three training pairs and 9 held-out cells, so the softmax has neither enough data nor enough structure to align confidence with accuracy. The certified transform gate then sidesteps the calibration question entirely: it produces a **unique** grammar-consistent prediction or no prediction, so the held-out top-1 is either 100% or the gate abstains. On the public ARC corpus, public ARC evidence surface improves from v10 eval 4 tasks/187 cells to v15_measured eval 7 tasks/1057 cells; v15_measured train remains 24 tasks/1187 cells. The hypothesis-induction layer described in Section 10 adds transform families beyond the older v10 grammar floor. Under the matched-baseline parity protocol `crlp.arc_induction.parity.v1`, on the synthetic move-to-anchor suite the baseline grammar certifies 0/4 tasks and the hypothesis-induction loader certifies 4/4 with zero regressions on the previously supported tasks; the same harness applied to the public ARC corpus is reported below as a scoped evidence surface rather than as a claim of competitive ARC performance.

8.4. Cross-domain port

To show the substrate is not ARC-specific, the same `NNHypothesis` + `NNProgram` + evidence schema is instantiated in the `program_synthesis` task on a `filter_then_reduce` template (stages `enumerate → filter → reduce`, slots `filter.predicate_id`, `reduce.kind`, `output.type`). The existing certificate checker for the list-DSL abstractions accepts the compiled programs unchanged, and `make program-synthesis` emits **nine certified `crlp.hypothesis.evidence.v1` rows** (one per abstraction in the library: `count_positive`, `count_negative`, `count_even`, `sum_positive`, `sum_negative`, `any_positive`, `all_positive`, `first_positive_or_zero`, `any_even`). Eight of the nine cover one task-instance event each; `any_even` certifies but covers zero on the current fixture-task pool — the row appears in `results/latest/summary.md` as a "Program Synthesis Hypothesis Evidence" table. A corresponding `prime_gap` hypothesis cross-port is not yet part of the supported evidence.

8.5. One shared scorer across all four tasks

The cross-task panels in Figure 7 trained a separate `f_theta` for each task. As an ablation, we replace the four scorers with **one shared scorer of the same architecture** — a single `[96 → 64 → 32 → 1]` MLP that takes a task-tagged action row (a fixed 96-dimensional envelope with the task id as a one-hot prefix) and softmaxes over each task's candidate set. The shared scorer is trained on a uniform-random draw across the four tasks at matched per-task update budget; for the same total budget we also train four single-task baselines. Numbers are means across three seeds: 16 epochs × 12 000 steps per seed, for 192 000 total updates, captured from `crlp.multitask.summary.v1` rows. **Three-seed steady state is what the paper claims; the single-seed snapshot from the earlier capture is retained in `paper/data.typ` as historical context only.**

Task	Cands	Shared top-1	Single top-1	Δ top-1 (3-seed)	Shared NLL	Single NLL
`prime_gap`	129	**14.1%**	13.6%	+0.4 pp	2.6648	2.6733
`grid_path`	4	90.9%	**91.1%**	−0.2 pp	0.3216	0.2999
`delayed_claim`	4	**98%**	97.9%	+0.1 pp	0.0472	0.0412
`arc_agi`	10	26.2%	**42.8%**	−16.6 pp	4.0827	8.2754

Table 7: Shared **one-MLP-on-all-four-tasks** vs **single-task** baseline at three-seed steady state, with 192 000 total updates per seed. Same `[96 → 64 → 32 → 1]` MLP, same task-tagged 96-feature envelope, same learning rate, three seeds, mean reported. At steady state the multi-seed picture is much quieter than the single-seed snapshot suggested: `delayed_claim` is statistically tied (+0.1pp); `prime_gap` and `grid_path` are tied within sub-1pp; **`arc_agi` keeps a 16.6pp single-task advantage** on the trained `0520fde7` task, down from the −28.6pp gap reported at the earlier 24 000-update budget. The "positive transfer on delayed_claim" framing from the one-seed snapshot did not survive replication and is retracted; the load-bearing multitask finding instead comes from the OOD evaluation in Table 8 below.

The load-bearing positive transfer signal is on OOD, not on the trained task. In the same-envelope OOD evaluation, we evaluate the same two ARC scorers (shared and single-task) — both trained only on `0520fde7` — on a held-out suite of 12 same-envelope synthetic ARC tasks (7×3 → 3×3 panel-composition tasks across families like `copy_left`, `copy_right`, `union`, `xor`, `row_any_intersection`). The shared scorer beats the single-task scorer on this OOD suite, even though it lost to it on the trained task in Table 7. This is consistent with the standard interpretation of single-task vs multi-task training: the single-task scorer is over-specialising to the 270 effective training rows of `0520fde7` (3 train pairs × 9 cells × 10 candidate colors) while the shared scorer is constrained by joint training on three unrelated tasks to find more transferable features.

Scorer	OOD top-1 (3-seed)	OOD NLL	OOD ECE
shared (multi-task)	**48.4%**	2.6278	0.4481
single-task ARC	42.8%	4.3723	0.482

Table 8: Same-envelope ARC out-of-distribution evaluation. Both scorers were trained only on `0520fde7`; this table evaluates each on a held-out suite of 12 synthetic 7×3 → 3×3 panel-composition tasks with the same 96-feature action envelope and the same 10-way color action space. Three seeds, aggregate row. **The shared multi-task scorer beats single-task by +5.57pp top-1 on OOD** despite losing by 16.6pp on the trained task. NLL drops by 40% under shared (2.63 vs 4.37), confirming that single-task is producing confidently wrong predictions on unseen tasks.

The reading is deliberately narrow. On the fixed ARC task, shared training loses in-distribution top-1; on same-envelope OOD, it improves top-1, NLL, and ECE. The certificate-gating contract makes that trade acceptable because the gate owns correctness where structure is already certified, while the scorer is most useful on the residual tasks the gate does not cover. Under a pure-scorer system this would be a direct memorization-vs-generalization tradeoff; under a verifier-gated system, the gate handles covered structure and the scorer can be optimized for residual ranking.

Two boundary checks keep the claim from becoming too broad. First, the calibration audit shows a real tradeoff: on fixed `0520fde7`, shared training lowers NLL (4.08 vs 8.28) while worsening ECE (0.6995 vs 0.5589); on same-envelope OOD, it improves both NLL (2.63 vs 4.37) and ECE (0.4481 vs 0.4820). Second, the delayed-claim leave-one-out row shows that transfer is sibling-specific rather than universal: all-domain training beats claim-only by +1.40pp, removing `grid_path` hurts most, and removing ARC slightly helps.

The capacity sweep rules out the simplest architectural explanation. Wider is not monotonically better: `[96→32→16→1]` is the closest shared ARC row to single-task (-1.23pp), while `[96→256→128→1]` only partially recovers the `64x32` gap. Delayed-claim stays positive or tied at every capacity. The remaining question is whether the current small nets are undertrained, saturated, or missing cumulative protocol state.

Hidden	Params	ARC shared	ARC single	ARC Δ	Claim Δ
`32x16`	3649	39.2%	40.4%	−1.23 pp	+0.43 pp
`64x32`	8321	26.2%	42.8%	−16.63 pp	+0.1 pp
`128x64`	20737	26.2%	40.4%	−14.27 pp	+0.13 pp
`256x128`	57857	34.5%	42.8%	−8.3 pp	+0.43 pp

Table 9: Fixed-task capacity sweep at the same three-seed, 192 000-update protocol. The ARC gap is **not** monotonic in trunk size: `32x16` is closest to single-task on fixed `0520fde7`, while `256x128` only partially recovers the `64x32` baseline. Delayed-claim remains positive or tied at every capacity.

The scorer-invariance audit fixes the boundary between learned scoring and certified ARC evidence. The public ARC transform gate and budgeted proposer path currently do not consume a multitask scorer checkpoint, so the comparison is an invariance audit over the scorer-independent path. Across 400 public train tasks and 400 public eval tasks, accepted rule sets, prediction hashes, loader summaries, and budgeted proposer aggregates were identical under the `shared_scorer_context` and `single_task_scorer_context` labels (`0` exact gate/loader mismatches and `0` proposer mismatches). This strengthens the verifier-correctness claim, but it also narrows the scorer claim: the audit does not show that the shared scorer improves public ARC search, only that the certified gate/proposer measurements are not contaminated by scorer choice in the current implementation.

8.6. Main artifact-feedback results

The two central result rows are the C5 artifact-feedback rows: rejected-candidate learning and certified hypothesis-library feedback. The rest of the claim table is boundary evidence: ARC is a measured evidence surface, the capacity sweep guards architecture interpretation, and Signal is application pressure under replay, shadow, and guarded live-active logging. Each row is evaluated against a registered result artifact before the PDF is built, so unsupported or stale required claims fail before they become prose. The only non-blocking negative row remains the fixed-budget prime proposer claim, which is deliberately carried as a limitation.

Claim	Family	Status	Required
C1.prime_gate_delta	C1	**supported**	yes
C2.prime_proposer_fixed_budget	C2	contradicted	no
C4.arc_ood_shared_vs_single	C4	**supported**	yes
C1.arc_gate_scorer_invariance	C1	**supported**	yes
C1.arc_public_evidence_surface	C1	**supported**	yes
C3.arc_ood_calibration_tradeoff	C3	**supported**	yes
C4.delayed_claim_sibling_transfer	C4	**supported**	yes
C4.multitask_capacity_tradeoff	C4	**supported**	yes
C5.rejected_candidate_reason_learning	C5	**supported**	yes
C5.hypothesis_library_promotion	C5	**supported**	yes
C7.tiny_net_plateau_stop_rule	C7	underpowered	no
C8.trace_determinism_audit	C8	**supported**	yes
C9.signal_flight_replay_control	C9	**supported**	yes
C9.signal_live_neural_active	C9	**supported**	yes

Table 10: Generated claim status matrix. Required claims fail `make paper-support` if their registered evidence is missing, stale, underpowered, or contradicted.

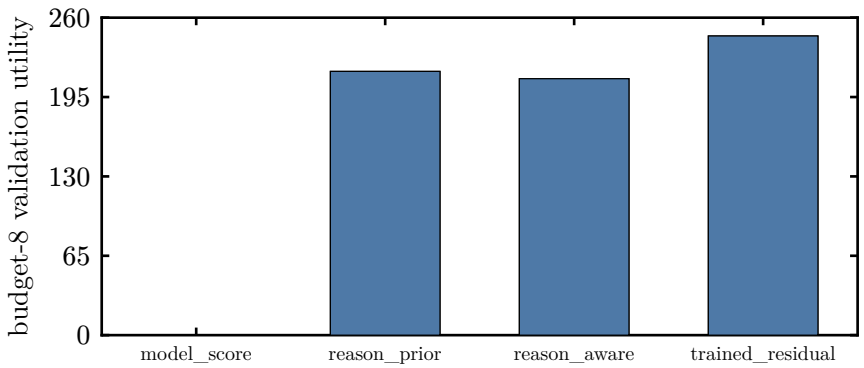


Figure 8: C5 rejected-candidate reason learning. reason learning repairs the prime proposer failure at budget 8: model_score selects 7 unsafe rows and causes 172 validation false rejections, while reason_prior reaches 6 accepted rules/216.0 utility and trained_residual reaches 7 accepted rules/245.0 utility with zero false rejections; prime/modular split sweeps pass, but delayed_claim has 0 mixed split rows.

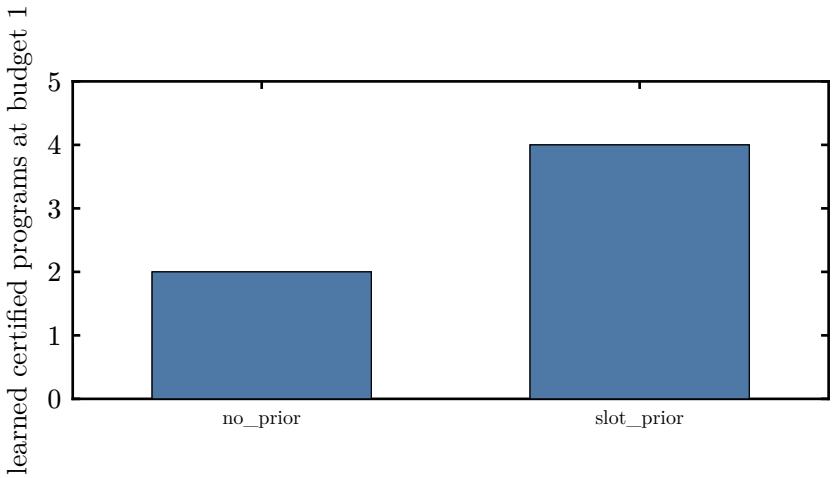


Figure 9: C5 certified hypothesis-library promotion. certified-library promotion supports fixture-level feedback: promoted move_to_anchor from 2 tasks/78 covered events with zero false rejections; slot priors raise learned certified@1 from 2 to 4 and utility@8 from 0.2119 to 0.5169. Held-out split feedback rows: 30.

The next rows are intentionally not headline performance claims. ARC is kept as an evidence surface: it exposes which transform and hypothesis families are certified, where the grammar misses, and whether scorer-independent gate measurements stay invariant. Signal is kept as application pressure: it exercises replay-certified ranking, shadow logging, and live neural-active execution without claiming that active neural control beats the heuristic.

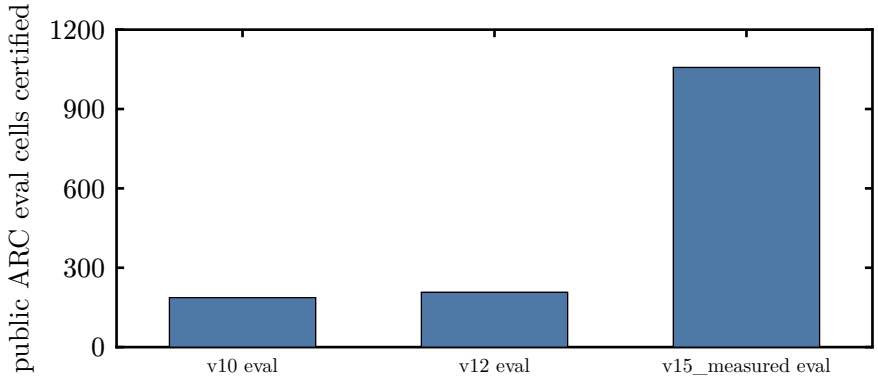


Figure 10: C1 public ARC evidence surface. public ARC evidence surface improves from v10 eval 4 tasks/187 cells to v15_measured eval 7 tasks/1057 cells; v15_measured train remains 24 tasks/1187 cells. The chart is intentionally scoped to certified covered cells, not overall ARC performance.

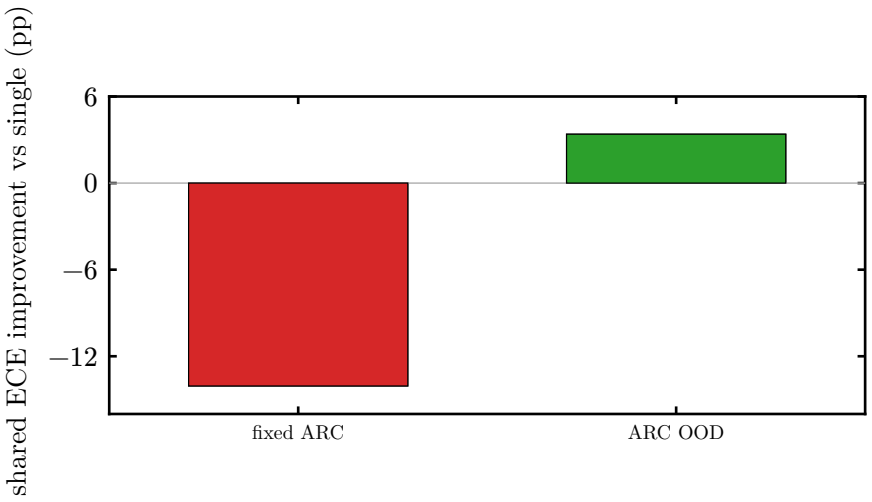


Figure 11: C3 ARC calibration tradeoff. Positive bars mean shared scoring has lower ECE than single-task scoring; negative bars mean worse ECE. ARC shared scorer is a calibration tradeoff: fixed-task NLL improves (4.0827 vs 8.2754) while ECE worsens (0.6995 vs 0.5589); OOD improves both NLL (2.6278 vs 4.3723) and ECE (0.4481 vs 0.4820).

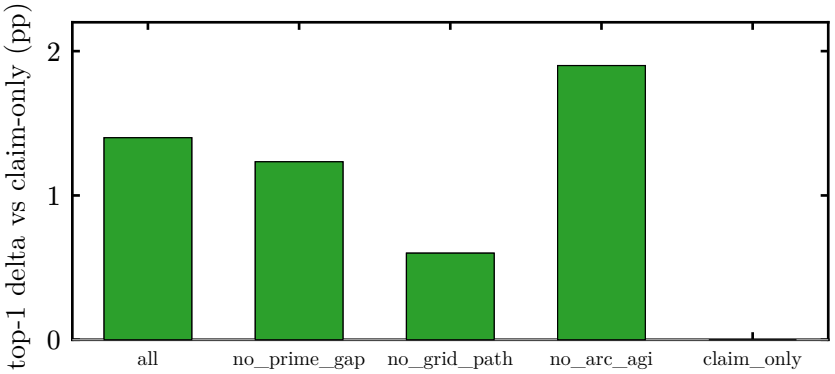


Figure 12: C4 delayed-claim sibling transfer. delayed_claim all-domain top-1 98.00% vs claim-only 96.60% (+1.40pp); removing grid_path falls to 97.20%, while removing ARC rises to 98.50%.

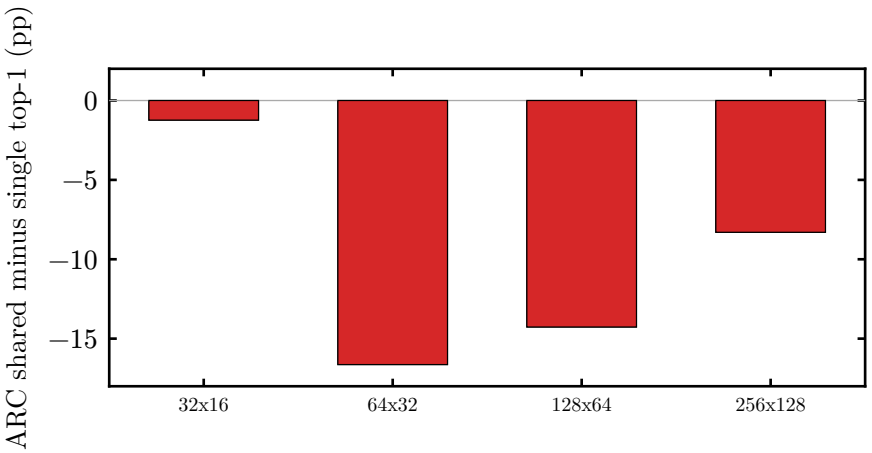


Figure 13: C4 capacity boundary. Negative bars mean the shared scorer still trails the single-task ARC scorer at that trunk size. capacity sweep rules out a simple wider-is-better ARC story: best fixed-ARC shared row is 32x16 at 39.20%, but every shared ARC row trails single-task; delayed_claim deltas remain +0.10pp to +0.43pp.

9. Application Pressure: Signal Flight

Signal is the practical reach test for the protocol, but the claim is intentionally narrow. The current result is not "a tiny net flies the game" and not "neural control beats the heuristic." Signal is split into generated evidence rows: an offline counterfactual replay row, a live shadow row, and a live neural-active row. In the replay row, each sampled state enumerates nine WASD candidates, the Signal simulator replays the candidates, and the ranker learns from replay-certified utility, damage, and death outcomes.

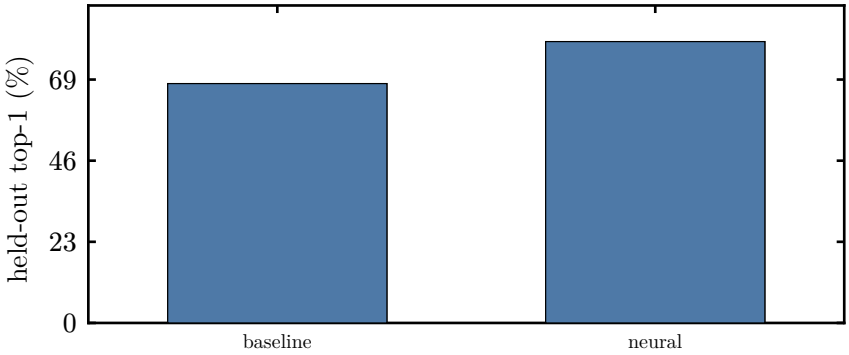


Figure 14: C9 Signal replay-control vignette. Signal offline replay ranker top-1 79.76% vs baseline 67.86% (+11.90pp), utility 0.1432 vs 0.1359, regret 0.0073 vs 0.0146, damage 2.38% vs 2.38%, deaths 0.00% vs 0.00%; live shadow frames=22191. The baseline is the primitive no-safety goal-heading WASD policy. This row supports bounded replay-certified ranking only; live execution is tracked by the separate neural-active claim.

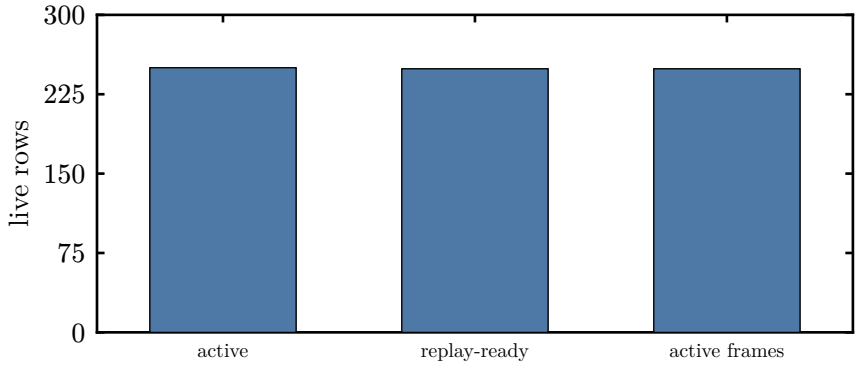


Figure 15: C9 Signal live neural-active execution. Signal live neural-active pilot executed 250 recent neural-source rows from checkpoint f12b644ed9fa with p95 inference 0.124 ms, 249 replay-ready rows, safety-gate rows=250, damage events=0, deaths=0; no baseline comparison is claimed. This row supports live guarded neural-active execution and replay-ready telemetry, not comparative utility or safety versus the heuristic baseline.

This is why Signal belongs in Table 3 even before it becomes a headline control experiment: it changes the environment class without changing the contract. Candidate enumeration is still explicit, the verifier is authoritative simulator replay or logged live telemetry, the certificate is a replayed or replay-ready outcome record rather than a theorem-like proof, and the neural component still ranks candidates rather than owning safety. The next unsupported gate is comparative active control: summarize active neural episodes against a heuristic baseline with utility, regret, damage, and death.

10. Artifact Feedback and Audit Contract

The hybrid architecture is the contribution; the audit contract is the discipline that makes the claims reproducible. Six pieces matter:

The generated C5 rows make the artifact-feedback loop explicit. Rejected candidate traces become reason priors and a trained residual that repair the prime/modular proposer economics without introducing false rejections (Figure 8). Repeated zero-false-rejection hypothesis bindings become a reusable library prior and feed that prior back into budgeted proposal ranking (Figure 9). In both cases, the method is the experiment: verified claim outcomes become the next round's proposal features, and the claim compiler decides whether the paper may say the loop helped.

- **Stable action IDs.** Trace rows carry `schema: "crlp.trace.v1"`, `trace_version: 1`, and `action_id_contract: "stable_unique_action_ids"`, plus a declared ranking policy and selected-score margin. Replay resolves `target_index` to the trace row's `action_id`, then maps that stable id back into the regenerated candidate row. The trace determinism audit projects rows onto replay-semantic fields, hashes that projection, and reports the count of `rank_fp_sensitive` decisions rather than requiring bit-identical logits.
- **Checkpoint identity.** SHA-256 over the exact checkpoint bytes, plus `parent_model_hash`, `feature_encoder_version`, and `feature_set` in metadata. Successor models declare their parent; comparison reports carry both hashes and the evaluated `feature_set`.
- **Hypothesis layer.** `NNHypothesis` (header `include/nn_hypothesis.h`) sits one level above `NNProgram`: a **template** declares a typed operator skeleton as a sequence of stages with named, typed slots; a **binding** fills each slot with a concrete value inferred from observed task evidence; `nn_hypothesis_compile` produces a composed `NNProgram` whose canonical id binds back to the hypothesis template, so the existing verifier and closed-loop policy treat it as any other program. `crlp.hypothesis.evidence.v1` rows are emitted by `nn_hypothesis_write_evidence_json`, a

library function reused by both `arc_agi` and `program_synthesis`. The provenance contract holds: state-observation slot values are `NN_FEATURE_STATE_OBSERVATION`, gate-derived support / residual gain appear only on the evidence row, never on the hypothesis feature vector that a selection-mode proposer scores against.

- **Certificate-gated baselines.** `crlp.arc_induction.parity.v1` records the matched baseline-vs-induction certification delta per split (training, evaluation) on the public ARC corpus or any synthetic split; `crlp.program_proposer.summary.v1` records matched-budget rule-set comparisons across `{random, grammar-order, structural, learned, closed-loop, oracle}`. Both schemas attest **what was held fixed** between rows, which is the load-bearing thing for cross-task comparison.
- **Pure C, no third-party deps.** The neural library is dense layers, sigmoid/ReLU/tanh/linear, MSE and softmax cross-entropy, SGD, batched 4-by-4 tiled forward/backward, cloned networks, action scoring, pairwise ranking loss, and cached policy-gradient updates with optional KL/entropy regularization. ≈3k lines for the core; the current certified modular gate implementation is split into focused C modules for data, features, rule search, training, evaluation, and replay.
- **CI-baseline runs.** A pinned `ubuntu-24.04` + `gcc-13` image runs `make check` and the ARC induction smoke on every push and PR, runs `make results` nightly, captures `crlp.environment.v1` for runner identity, and emits a **timing-stripped** `comparison.canonical.jsonl` so the diff that gates "should the paper rewrite?" cannot be triggered by wallclock noise. Laptop runs produce the same artifacts; drift shows up as a visible `cc_version` or `git_dirty` difference rather than as silent numeric drift.

11. Limitations

L1. **Experiment-family scope is frozen for this paper.** The current evidence matrix is already broad enough to support the protocol claim. Additional domains would make the draft less falsifiable unless they replace a weaker row or pass the same candidate/verifier/gate/scorer trace standard. New domains belong in roadmap notes, not body evidence.

L2. **Proposer learning remains weak (Axis 2).** The fixed-budget baseline shows grammar order and a simple structural heuristic beat the current neural proposer on both the unconditional candidate grammar and the conditional gap grammar. The load-bearing claim is certificate-gated rule discovery, not learned proposer superiority. The negative result is reported deliberately.

L3. **Cross-task verifier-RL comparison missing (Axis 3).** Calibration evidence is on the prime-gap modular gate instance only. The pipeline supports running strict-vs-KL on grid-path and delayed-claim; the comparison is not in canonical results yet.

L4. **Limited cross-domain gate coverage (Axis 1).** The prime-gap modular gate instance and grid-path have the strongest **gate ablations** in the strict sense. Delayed-claim now has a production threshold+margin gate that resolves its held-out deterministic generator, including the hard `trust` versus `request_more_evidence` residual, and it now reports shifted/noisy stress rows. Those rows are mixed: the direct grammar is exact under clean seed shift but false-rejects under observation noise, while the indirect proxy avoids false rejections but leaves a larger residual. This is evidence of the robustness frontier, not a broad noisy-claim solution. ARC-AGI's "gate" is a different architecture (transform grammar) over the same task, not the same scorer with vs without filtering, so it is reported separately in Figure 6.

L5. **Statistical rigor.** Default `make prime-gap-experiments` runs 1 seed; the harness supports more (`PRIME_GAP_EXPERIMENT_SEEDS=…`). The 9.13 → 45.12 → 68.87% deltas, the 73.22% baseline saving, the grid-path 75.80 -> 90.10 cheap-bound gate delta, the 95.30 mixed-gate upper bound, and the modular-rule-search negative result all need confidence intervals across ≥ 5 seeds before any of them should be treated as a stable property.

L6. **Learned-wheel calibration tradeoff** needs a reliability-diagram pass. The learned wheel improves ranking sharply (Table 4) but has higher ECE (0.1402) than the raw minimal row (0.1076).

L7. **ARC ECE outlier.** The 0.5574 ECE on the `0520fde7` cell scorer is reported, with the structural explanation (10 colors × 3 train pairs is too sparse for the softmax to align with accuracy). It is not yet replicated across multiple ARC tasks; the structural argument is suggestive, not proof.

L8. **Signal lacks a comparative active-control result.** The Signal rows now support offline replay ranking, live shadow evidence, and guarded live neural-active execution (Figure 15). They do not yet support the stronger claim that active neural control improves utility, regret, damage, or death versus the heuristic baseline.

L9. **No competitive large-system baseline.** The related-work comparison now positions the protocol against learned verifiers, RLVR, AlphaGeometry-style formal search, DreamCoder, Hypothesis Search, and ARC systems, but the artifact does not empirically compare against those large systems. That is acceptable for the protocol claim; it would not be acceptable for a claim of competitive theorem proving, language-model reasoning, or ARC performance.

L10. **Single-runner numbers.** All headline percentages were captured from a single CI baseline run (`ubuntu-24.04` + `gcc-13`, see `results/latest/environment.json`). Laptop runs produce different `model_ms_per_sample` and minor numerical differences in `nll` / `ece` at the third decimal; the canonical pipeline strips timing fields and now exposes a replay-semantic trace hash plus `rank_fp_sensitive` counts, but a cross-machine double-blind has not yet been run. Raw logits remain diagnostics rather than semantic trace identity.

Claim	Status	Generated limitation candidate
`C2.prime_proposer_fixed_budget`	contradicted	static neural proposer trails deterministic baselines on at least one grammar; candidate static=72.49 saved vs grammar=73.22, conditional static=2.15 saved vs grammar=55.57.
`C7.tiny_net_plateau_stop_rule`	underpowered	tiny-net plateau audit covers 4 capacities across 4 domains; 10/16 shared rows meet the plateau threshold, and ARC shared width verdict picks 16x8 as best hidden.

Table 11: Generated limitation candidates from the paper-claim compiler. The required claim support gate is currently passing. Optional contradicted claims remain visible here instead of blocking PDF compilation.

12. Conclusion

The paper's strongest claim is narrow: certified candidate mechanisms should own admissibility, and small learned components should rank, calibrate, reuse, or learn around those certified hypotheses. The two main empirical contributions are the feedback loops. Rejected candidate rows become reason-coded targets for a later safety-first residual ranker; certified hypotheses become reusable library priors and are checked on held-out proposal budgets. The gate, tiny-net, ARC, and Signal rows matter because they keep that claim honest: modular gates show the protocol can sharply reduce residual search, capacity audits block a naive wider-is-better story, ARC exposes priors and misses rather than claiming broad competence, and Signal keeps the loop under replay-grounded application pressure without claiming comparative live-control performance.

The negative fixed-budget proposer result is therefore not an embarrassment. It is evidence that the protocol is falsifiable: a flat neural proposer over structural rows is weak, but the verifier trace can

still become useful training signal. The broader implication is pragmatic. When correctness, safety, or auditability matter, end-to-end learning should not own the whole decision. Verifier-grounded ranking substrates provide a smaller, auditable contract: symbolic or replay checkers own admissibility, neural components spend capacity on search and prioritization, and the trace artifacts make both success and failure reusable.

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